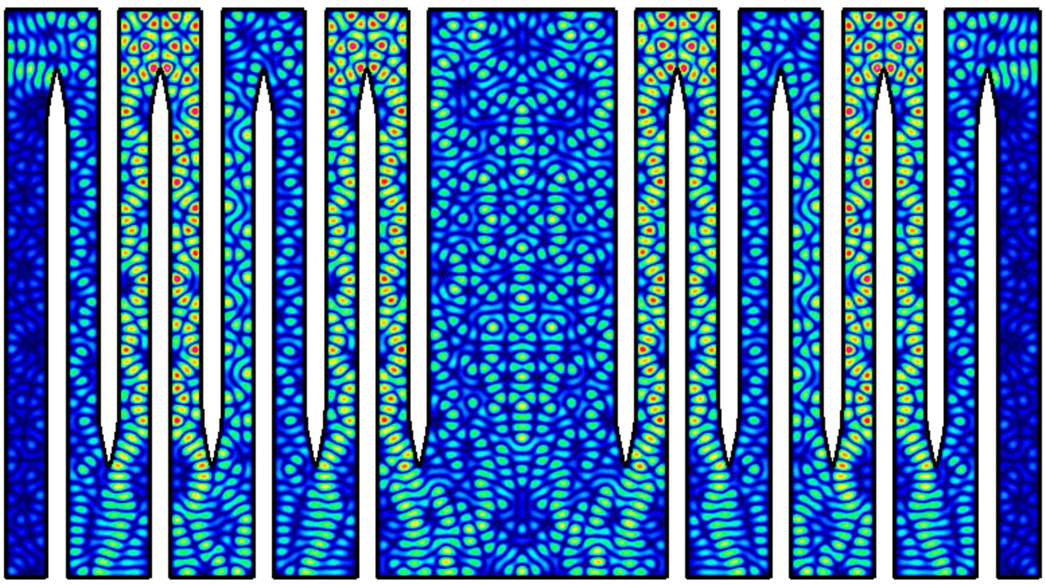
Waves 2026 - Program

The 17th International Conference on Mathematical and Numerical Aspects of Wave Propagation

John Molson school of business Concordia University in Montreal

Canada, June 22-26 2026



Sunday - Thursday

Registration open (times TBD)

Monday

Welcome · 08:30-09:00

Plenary: Guillaume Bal

09:00-10:00 · Session Chair: Chrysoula Tsogka

Coffee 10:00-10:30

Block A · 10:30–12:30

T1 — Helmholtz #1 Session - Chair: Nilima Nigam

Time	Title	Authors
10:30	Convergence of the WaveHoltz iteration on R^d	**Elliot Backman**, Olof Runborg
11:00	Effective models for Helmholtz's equation in a corner with a thin layer	**Cédric Baudet**, Sonia Fliss, Patrick Joly
11:30	The largest ever Helmholtz simulations in inhomogeneous media	**Max Cubillos**, Edwin Jimenez
12:00	Nonlinear acoustic inverse problem with cross-correlation data for passive imaging	**Jean Dutheil**, Florian Faucher

T2 — Boundary Element Methods & Integral Equations #1 Session - Chair: Marc Bonnet

Time	Title	Authors
10:30	Efficient evaluation of the two-layered acoustic Green's function	**Stefan A. Sauter**, Laura Depedrini, Benedikt Grässle, J. Markus Melenk
11:00	Fast high-order solvers for the Lippmann-Schwinger equation in piecewise-smooth heterogeneous media	**Thomas G. Anderson**, Juan Burbano-Gallegos, Luiz M. Faria, Carlos Pérez-Arancibia
11:30	A fast direct solver based on FEM-BEM coupling for time-harmonic electromagnetic scattering	**Johann Bourhis**, Emanuele Arcese, Francis Collino
12:00	Truncated kernel windowed Fourier projection: a fast algorithm for the 3D free-space wave equation	**Nour G. Al Hassanieh**, Alex H. Barnett, Leslie Greengard

T3 — Absorbing Boundary Conditions #1 - Session Chair: Maryna Kachanovska

Time	Title	Authors
10:30	Revisiting Artificial Boundary Conditions Through Modal Expansion	**Hélène Barucq**, Florian Faucher, Olivier Lafitte, Ha Pham
11:00	Transparent boundary conditions in heterogeneous electromagnetic waveguides	**Anne-Sophie Bonnet-Ben Dhia**, Lucas Chesnel, Sonia Fliss, Eric Lunéville, Aurélien Parigaux
11:30	Domain Truncation for Quasilinear Wave Equations	**Thomas Hagstrom**
12:00	Finite element discretizations of hyperboloidal compactification layers for unbounded time-domain wave problems	**Markus Wess**, Anıl Zenginoğlu

T4 — Acoustic Wave Propagation & Scattering #1 Session - Chair: Christophe Hazard

Time	Title	Authors
10:30	Time dependent acoustic scattering from penetrable objects with nonlinearities	**Paschalis Angelidis**, Alexander Rieder
11:00	Analysis of the geometric spectral properties of electromagnetic waveguides	Philippe Briet, **Maxence Cassier**, Thomas Ourmières-Bonafos, Michele Zaccaron
11:30	Rendering disordered waveguides transparent to waves	**Matthieu Davy**, Isam Ben Soltane, Daniel Lenz, Stefan Rotter
12:00	Spectral properties of a kinetic equation coupled with the linear wave system	**Bruno Després**

T5 — Inverse Problems & Imaging #1 - Session Chair: Shixu Meng

Time	Title	Authors
10:30	Recovery of acoustic properties in the Westervelt equation using high-frequency large-amplitude waves	**Sebastian Acosta**, Benjamin Palacios
11:00	A neural operator framework for solving inverse scattering problems	**Victor Chenu**, Houssem Haddar, Hadrien Montanelli
11:30	A deep learning approach for optical cloaking device design	**Elsie Cortes**, Camille Carvalho, Symeon Papadimitropoulos, Chrysoula Tsogka
12:00	A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media	**General Ozochiawaeze**, Rafael Ceja Ayala, Isaac Harris

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Lunch 12:30-14:00

Plenary: Christophe Hazard

14:00-15:00 · Session Chair: Anne-Sophie Bonnet-Ben Dhia

Coffee 15:00-15:30

Block B · 15:30–17:30

T1 — Electromagnetic Waves & Maxwell Equations #1 - Session Chair: Carlos Perez-Arancibia

Time	Title	Authors
15:30	Optimization of EM-Chiral Scatterers	**Tilo Arens**, Marvin Knöller, Raphael Schurr
16:00	Domain derivative based shape reconstruction for time-dependent Maxwell's equations	**Marvin Knöller**
16:30	Towards fast conservative methods for the numerical resolution of Maxwell's equations in multipatch domains	**Ángel Pita-da-Veiga**, Jerónimo Rodríguez, Rafael Vázquez
17:00	The Hermite-Taylor Discrete Correction Function Method for Surface-Driven Electromagnetic Problems	**Yann-Meing Law**

T2 — Elastic Waves & Elastodynamics #1 - Session Chair: Stephanie Chaillat

Time	Title	Authors
15:30	Mathematical and numerical modelling of elastic wave propagation with smooth nonlinear contact conditions	**Alexandre Imperiale**, Bruno Lombard, Sébastien Imperiale
16:00	Green tensor for strain gradient elastodynamics	**Marc Bonnet**
16:30	Reconstruction of the Shear Modulus in Nearly Incompressible Media via Adaptive Spectral Inversion	**Nagham Chibli**, Martin Genet, Sébastien Imperiale
17:00	Spectral properties of waves in high-contrast media	Yuri A. Godin, Leonid Koralov, **Boris Vainberg**

T3 — Helioseismology & Astrophysics Session Chair: Raphael Terrine

Time	Title	Authors
15:30	Reconstructing inertial-mode flows from the correlation function of acoustic modes	Hélène Barucq, Florian Faucher, **Damien Fournier**, Laurent Gizon, Zhi-Chao Liang, Ha Pham
16:00	Uniqueness results in time-harmonic passive imaging with applications to helioseismology	**Thorsten Hohage**, Björn Müller
16:30	Asymptotic Solutions to the Radial Wave Equation in Kerr Spacetime	**John Levis**, Thomas Hagstrom
17:00	Stability of a passive inverse parameter problem with applications to helioseismology	**Björn Müller**, Thorsten Hohage

T4 — Water Waves & Wave-Structure Interaction #1 - Session Chair: Olivier Lafitte

Time	Title	Authors
15:30	Continuous and discrete stability of dispersive perturbation of hyperbolic initial boundary value problems	**Geoffrey Beck**, Nicolas Crouseilles, Ludovic Martaud
16:00	The return to the equilibrium of a floating cylinder in the Boussinesq regime	Geoffrey Beck, **Ewan Contentin**, Ludovic Martaud
16:30	A Theory for Three-Dimensional Wave Breaking Based on Crest Instability of Short-Crested Stokes Waves	**Frédéric Dias**, Ayoub Mansar, Matt Turner, Tom Bridges
17:00	Wave-Based Stability Analysis of Couette and Gravity-Driven Bilayer Film Flows	**M. Al Ahmadi Hammou**, M. Scholle, P. H. Gaskell

T5 — Nonlinear Waves, Solitons & Dynamical Systems - Session Chair: Behrooz Yousefzadeh

Time	Title	Authors
15:30	Localized Time-Periodic Patterns in a 1D Lattice	**Jason J. Bramburger**, Eric Berglund, Björn Sandstede
16:00	Bistability of travelling waves and mesa states in a mass-conserved reaction-diffusion system	**Jack M. Hughes**, Saar O. Modai, Leah Edelstein-Keshet, Arik Yochelis
16:30	Leafnet algorithm for N-dimensional nonlinear hyperbolic systems	**Emmanuel Lorin**, Arian Novruzi
17:00	Exploring Explicit Formulas and Multi-solitons for Integrable Systems	**Ruoci Sun**

Tuesday

Plenary: Barbara Romanowicz

09:00-10:00 · Session Chair: Serge Prudhomme

Coffee 10:00-10:30

Block A · 10:30–12:30

T1 — Helmholtz #2 - Session Chair: Thomas Anderson

Time	Title	Authors
10:30	The Helmholtz boundary element method suffers from the pollution effect	**María Ignacia Fierro**, Jeffrey Galkowski, Manas Rachh, Euan Spence
11:00	A Trefftz Continuous Galerkin method for Helmholtz problems	**Nicola Galante**, Bruno Després, Emile Parolin
11:30	High-order frequency derivatives of Helmholtz BIE solutions: fast computation and applications	**Hiroshi Isakari**
12:00	A Helmholtz Solver for Open Domains with Complex Geometry using WaveHoltz and Overset Grids	Jeffrey Banks, William Henshaw, **Ngan Le**, Donald Schwendeman

T2 — Boundary Element Methods & Integral Equations #2 - Session Chair: Stefan Sauter

Time	Title	Authors
10:30	Efficient 3D elastodynamic multiple scattering using the Method of Reflections	**Stéphanie Chaillat**, Marion Darbas
11:00	Fast Singular-Kernel Convolution on General Non-Smooth Domains via Truncated Fourier Filtering	**Jinghao Cao**, Oscar Bruno
11:30	Convergence analysis for Numerical Steepest Descent contour deformation methods	**Thomas Caussade**, David P. Hewett
12:00	Space-time BEM based coupling strategies for the analysis of infrastructures solicited by seismic waves	**Giulia Di Credico**, Alessandra Aimi, Heiko Gimperlein, Chiara Guardasoni, Francesco Freddi, Roberto Valentino

T3 — Absorbing Boundary Conditions #2 - Session chair: Thorsten Hohage

Time	Title	Authors
10:30	Transparent boundary conditions for the stellar oscillation equations without Cowling approximation	**Janosch Preuss**, Hélène Barucq, Florian Faucher, Ha Pham, Sébastien Tordeux
11:00	Interfacial Radiation Boundary Conditions for Topologically Protected Edge Modes	**Mason McCallum**, Thomas Hagstrom
11:30	Complex scaling for the homogeneous wave equation with scaling direction non-normal to the transparent boundary	**Lothar Nannen**, Constantin Pierer, Markus Wess
12:00	A potential theory on weighted graphs	**Fernando Guevara Vasquez**, Trent DeGiovanni

T4 — Acoustic Wave Propagation & Scattering #2 - Session Chair: Frederic Dias

Time	Title	Authors
10:30	Scattering problems in junctions of stratified media	Sarah Al Humaikani, Anne-Sophie Bonnet-Ben Dhia, **Sonia Fliss**
11:00	Spectral analysis of a closed perturbed waveguide filled with a hyperbolic medium	Maryna Kachanovska, **Dylan Machado**
11:30	Exceptional Points and Defective Resonances in a Sound-Soft Scattering Problem	**Kei Matsushima**
12:00	Flow Rate Measurement in a Heterogeneous Fluid with Acoustic Waves	**Alexander McSweeney-Davis**, Lorenzo Audibert, Houssem Haddar

T5 — Inverse Problems & Imaging #2 - Session Chair: Max Cubillos

Time	Title	Authors
10:30	Explicit inverse scattering for the one-dimensional Schrödinger equation	**Peter Gibson**
11:00	On the role of regularisation parameters in Adaptive Eigenspace Inversion	**Marie Graff**, Nasrin Nikbakht, Melissa Tacy
11:30	The Inverse Elasto-Acoustic Problem	**Patrick Grice**, Yassine Boubendir, Zoi-Heleni Michalopoulou
12:00	Adapting Full-Waveform Inversion for Subsurface CO_2 Sequestration	**Alison Malcolm**, Vahid Entezaar-Saadat, Xingda Jiang, Abolfazl Khan Mohammadi, Colin Farquharson

Lunch 12:30-14:00

Plenary: Leonardo Zepeda-Núñez

14:00-15:00 · Session Chair: Behrooz Yousefzadeh

Coffee 14:00-15:30

Tue · Block B · 15:30–17:30

T1 — Electromagnetic Waves & Maxwell Equations #2 - Session Chair: Tilo Arens

Time	Title	Authors
15:30	A Stochastic Boundary Element Method for Electromagnetic Scattering in Random Media	**Titouan Marquaille**, Thomas Bonnafont, Arnaud Coatanhay
16:00	A Discontinuous Petrov–Galerkin Framework for Time-Harmonic Maxwell Equations in Microwave-Heated Flows	**Oreste Marquis**, Bruno Blais, Matthias Maier
16:30	Computational spectral geometry for the Maxwell cavity problem	**Nilima Nigam**, Kewen Bu, Jayda Grant, Rodrigo Stehling
17:00	A Hermite-Taylor Compact Discrete Correction Function Method for Maxwell's Equations	**Seth Malonson**, Yann-Meing Law, Jean-Christophe Nave

T2 — Elastic Waves & Elastodynamics #2 - Session Chair: Barbara Romanowicz

Time	Title	Authors
15:30	Comparison of numerical coupling strategies in time-harmonic elasto-acoustic simulations	**Pierre Dubois**, Hélène Barucq, Florian Faucher, Isabelle Ramière, Raphaël Prat
16:00	A well-posed effective strain-gradient model for elastic waves in arbitrary periodic media	**Nicolas Auffray**, Marc Bonnet, Rémi Cornaggia, Giuseppe Rosi
16:30	Mathematical models for wave propagation phenomena generated by landslides	Juliette Dubois, Natacha Guegan-Fau, **Sébastien Imperiale**, Anne Mangeney, Jacques Sainte-Marie
17:00	High-order Simulation of Elasto-Acoustic Wave Propagation Using Hybrid Nonconforming Methods (HHO)	**Romain Mottier**, Alexandre Ern, Laurent Guillot

T3 — Periodic Media, Metamaterials & Homogenization #1 - Session Chair: Svitlana Mayboroda

Time	Title	Authors
15:30	Diffraction by a rough thin layer on an arbitrary shaped object: the periodic and random cases	**Pierre Boulogne**, Sonia Fliss, Laure Giovangigli, Justine Labat
16:00	High-order stochastic homogenization of the Helmholtz transmission problem	**Laure Giovangigli**
16:30	Wave propagation and gaps in three-dimensional periodic media	**Yuri Godin**, Boris Vainberg
17:00	On the surface Bloch waves in truncated periodic media	**Bojan B Guzina**, Shixu Meng, Prasanna Salasiya, Long Nguyen

T4 — Time-Domain Methods & A Posteriori Estimation #1 - Session Chair: Marcus Grote

Time	Title	Authors
15:30	A note on the discrete Unmapped Tent Pitching for the heterogeneous wave equation	**Marcella Bonazzoli**, Gabriele Ciaramella, Ilario Mazzieri
16:00	Decoupling Wave-Heat-Type Problems	**Julian Dörner**, Benjamin Dörich
16:30	Mixed Precision Implicit Runge–Kutta and Explicit Two-Derivative Runge–Kutta Methods	**Sigal Gottlieb**, Zachary J. Grant, César Herrera
17:00	Recovery-Based Error Indicator for Finite Difference Methods	**Ferhat Sindy**, Annalisa Buffa, Marco Picasso

T5 — Stochastic Propagation - Session Chair: Emmanuel Lorin

Time	Title	Authors
15:30	Effective attenuation and Gaussian statistics of the speckle for the paraxial wave equation	**Christophe Gomez**, Olivier Pinaud
16:00	Multiscale analysis of wave propagation in the branched flow regime	Josselin Garnier, **Natacha Guegan-Fau**, Sébastien Imperiale
16:30	Optical wave reconstruction in highly turbulent random media	**Kevin Luna**, Max Cubillos
17:00	Destabilization of traveling waves in a nonlinear lattice with nonreciprocal coupling	Andrus Giraldo, Stefan Ruschel, **Behrooz Yousefzadeh**

Thursday

Plenary: Andrea Moiola

09:00-10:00 · Session Chair Lise-Marie Imbert-Gerard

Coffee 10:00-10:30

Thu · Block A · 10:30–12:30

T1 — Helmholtz #3 - Session Chair: Sonia Fliss

Time	Title	Authors
10:30	The hp-FEM Does Not Suffer from the Pollution Effect When Applied to the Helmholtz Equation with Piecewise Gevrey Coefficients	Jeffrey Galkowski, **Mostafa Meliani**, Euan Spence
11:00	High-order Wavelet Galerkin Methods for Scattering Problems	**Michelle Michelle**, Bin Han
11:30	A HDG method with transmission variables for time-harmonic wave propagation problems	**Simone Pescuma**, Gwénaël Gabard, Théophile Chaumont-Frelet, Axel Modave
12:00	Scalable WaveHoltz in MFEM	**Amit Rotem**, Daniel Appelö, Will Pazner

T2 — Trefftz & Plane Wave Methods - Session Chair: Bruno Despres

Time	Title	Authors
10:30	A Model Reduction Approach for Wave Phenomena using dG-Trefftz Methods	**Tobias Born**, Karsten Urban
11:00	Preconditioning for the DG plane wave method	**Joseph Coyle**, Nilima Nigam
11:30	Polynomial quasi-Trefftz spaces for scalar linear equations	**Lise-Marie Imbert-Gérard**
12:00	Quasi-Trefftz spaces for time-harmonic Maxwell's equations	**Ilaria Fontana**, Lise-Marie Imbert-Gérard

T3 — Boundary Element Methods & Integral Equations #3 - Session Chair: Johannes Tausch

Time	Title	Authors
10:30	A posteriori estimates and adaptive boundary elements for the wave equation	**Heiko Gimperlein**
11:00	Towards kernel-free boundary integral methods for variable coefficients	**Benedikt Gräßle**, Stefan A. Sauter
11:30	Geometry-conforming Galerkin methods for the Lippmann-Schwinger equation on fractal domains	**David P. Hewett**, Joshua Bannister, Andrew Gibbs
12:00	An Unconditionally Stable Energetic Galerkin Scheme for Acoustic Penetrable Objects Using Time-Domain Surface Integral Equations	**Junze Shao**, Rui Chen, Hakan Bagci

T4 — Acoustic Wave Propagation & Scattering #3 - Session Chair: Lothar Nannen

Time	Title	Authors
10:30	Imaging in two-dimensional waveguides with complex geometry	**Gregory Mwamba**, Chrysoula Tsogka, Symeon Papadimitropoulos
11:00	The cut-off resolvent can grow arbitrarily fast in obstacle scattering	**Siavash Sadeghi**, Simon N. Chandler-Wilde
11:30	Application of the multivariate Loewner Framework to the modelling of acoustic propagation in rigid-frame porous materials	**Lucie Schwoebel**, Charles Poussot-Vassal, Denis Matignon, Rémi Roncen, Estelle Piot, Pierre Vuillemin
12:00	Optimization-based transionospheric SAR autofocus with screen projection precursor	Mikhail Gilman, **Semyon Tsynkov**

T5 — Inverse Problems & Imaging #3 - Session Chair: Isaac Harris

Time	Title	Authors
10:30	Transmission eigenvalues for biharmonic plate scattering: existence, discreteness, far-field recovery, and numerical computation	**Andreas Kleefeld**, Isaac Harris, Heejin Lee
11:00	Characterizing the Imaging Indicator of the Linear Sampling Method for the Inverse Medium Scattering Problem	Lorenzo Audibert, **Shixu Meng**
11:30	Spectrally characterizing subsurface targets with GPR-SAR	Arnold D. Kim, Symeon Papadimitropoulos, **Joseph Simpson**, Chrysoula Tsogka
12:00	Identifying defective units in infinite periodic arrays of point sources	**Dinh-Liem Nguyen**, Nhung H. Nguyen, Thi-Phong Nguyen

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Lunch 12:30-14:00

Plenary: Lilia Krivodonova

14:00 – 15:00 · Session Chair: Tom Hagstrom

Coffee 14:00-15:30

Thu · Block B · 15:30 – 17:30

T1 — Electromagnetic Waves & Maxwell Equations #3 - Session Chair: Yann-Meing Law

Time	Title	Authors
15:30	Broken-FEEC on multipatch domains with local refinements	**Frederik Schnack**, Martin Campos Pinto
16:00	Topological solutions and accelerated waves in electromagnetic theory	Radmila Sazdanovic, Paul Spears, **Semyon Tsynkov**
16:30	Energy-preserving simulation of electromagnetic waves at oblique incidence using Split-Field FETD method	**Valentin Wasquel**, Paula Aguilera, Michel Fournié, Denis Matignon
17:00	Derivation of polarization dependent interface conditions for Maxwell's equations	**David Wiedemann**, Ben Schweizer, Bérangère Delourme

T2 — Periodic Media, Metamaterials & Homogenization #2 - Session Chair: Sebastien Imperiale

Time	Title	Authors
15:30	Analysis of Wave Propagation in Time-Dependent Media	**Patrick Joly**, Marie Touboul
16:00	Homogenization of wave equations and dispersive profiles for ring solutions	**Ben Schweizer**, Grégoire Allaire, Agnes Lamacz
16:30	Radiation Boundary Conditions with Rational Function Approximations	**Steve Li**, Thomas Hagstrom
17:00	Error Estimation for a Homogenized Model of a Space-Time Modulated Metamaterial	**Jean-Francois Mercier**, Marie Touboul

T3 — Time-Domain Methods & A Posteriori Estimation #2 - Session Chair: Sigal Gottlieb

Time	Title	Authors
15:30	Adaptive FEM with explicit time integration for the wave equation	**Marcus J. Grote**, Omar Lakkis, Carina S. Santos
16:00	High-order contour-integral methods for strongly continuous semigroups	**Andrew Horning**, Adam R. Gerlach
16:30	Reliable and locally space-time efficient a posteriori estimates for the transient wave equation	**Nicolas Hugot**, Alexandre Imperiale, Martin Vohralík
17:00	A parallel space-time p-adaptive discontinuous Galerkin method for nonlinear acoustics	**Pascal Lehner**, Christian Wieners, Danielle Corallo

T4 — Water Waves & Wave-Structure Interaction #2 - Session Chair: Helene Barucq

Time	Title	Authors
15:30	Waves and transport in a circular water wave resonator: experiments and simulations	**Uwe Harlander**, Franz-Theo Schön, Ion D. Borcia, Rodica Borcia, Seymen Ilke Kaykanat, Michael Bestehorn
16:00	Propagation of the heave in a harbor: geometrical optics approach	**Olivier Lafitte**, Emmanuel Audusse, Catherine Sulem
16:30	Interactions between water waves and floating structures, a numerical method	Geoffrey Beck, **Alan Riquier**
17:00	Viscoelastic gravity wave force on a circular cylinder in open channel	**Boyuan Yu**, Vincent H. Chu

T5 — Helmholtz / Trefftz / BEM Solver Extensions - Session Chair: Andrea Moiola

Time	Title	Authors
15:30	Infinity as a finite boundary for Helmholtz equations	**Anıl Zenginoğlu**, Markus Wess
16:00	Quasi-Trefftz Discontinuous Galerkin Method for the Convected Helmholtz Equation with Smooth Coefficients	**Andréa Lagardère**, Lise-Marie Imbert-Gerard, Guillaume Sylvand, Sébastien Tordeux
16:30	Variational Quasi-Trefftz Method for Time-Harmonic First-Order Wave Propagation Systems	Matthias Rivet, Sébastien Pernet, **Sébastien Tordeux**
17:00	Fast 3D Time Domain Boundary Element Approach for Elastodynamics based on Multivariate Adaptive Cross Approximation	Vibudha Lakshmi Keshava, **Martin Schanz**

Friday

Plenary: Isaac Harris

09:00-10:00 · Session Chair: Daniel Appelo

Coffee 10:00-10:30

Fri · Block A · 10:30 – 12:30

T1 — Boundary Element Methods & Integral Equations #4 - Session Chair: David Hewitt

Time	Title	Authors
10:30	Maxwell à la Helmholtz: Frequency-Robust Direct Boundary Integral Equations for 3D PEC Scattering	**Carlos Pérez-Arancibia**, Catalin Turc
11:00	Fast assembly of H-matrix and FMM-based boundary operators for time-harmonic Maxwell equations	**Clément Aumonier**, David Levadoux
11:30	High-order Nyström method for scattering by penetrable inhomogeneous media with discontinuous refractive index	**Ambuj Pandey**, Krishna Yamanappa Poojara
12:00	Model reduction for a wave propagation problem in a slotted structure	**Augustin Leclerc**, Pierre Benjamin, Luc Giraud, Sofiane Haddad, Paul Mycek, Guillaume Sylvand, Isabelle Terrasse

T2 — Acoustic Wave Propagation & Scattering #4 - Session Chair: Lilia Krivodonova

Time	Title	Authors
10:30	Passivity enforcing multipole based method for time domain modelling of acoustic wave propagation in rigid frame porous media	**Lucie Schwoebel**, Denis Matignon, Remi Roncen, Estelle Piot
11:00	Quantitative ultrasound imaging in layered media	Josselin Garnier, Laure Giovangigli, Pierre Millien, **Sofia Suarez Casabiell**
11:30	Addressing Seismic Acquisition Repeatability in CO_2 Storage Monitoring Using Machine Learning	**Manon Sarrouilhe**, Mauricio Araya-Polo, Hélène Barucq, Henri Calandra, Stefano Frambati
12:00	Towards a Scale-Bridging Time Discretization Scheme for the Acoustic Wave Equation	**Dustin Mühlhäuser**, Roland Maier

T3 — Elastic Waves & Elastodynamics #3 - Session Chair: Semyon Tsynkov

Time	Title	Authors
10:30	Boundary determination in anisotropic elasticity	Joonas Ilmavirta, Mikko Salo, **Hjørdis Schlüter**, Daniel Windisch
11:00	Seismic Reflection of Three-Dimensional Plane Waves in Functionally Graded Triclinic Media with Interfacial Imperfections	**Akanksha Srivastava**, Federico J Sabina
11:30	An inverse problem for estimating effective parameters in asymptotic elastodynamics with non-periodic roughness	**Rodrigo Zelada**, Alexandre Imperiale, Philippe Moireau
12:00	Exponential and harmonic regimes of anti-plane shear wave in a piezoelectric thin layer over half-space	**Anna Zemlyanova**, Gennadi Mikhasev, Arindam Nath, Jia Fei

T4 — Inverse Problems & Imaging #4 - Session Chair: Andreas Kleefeld

Time	Title	Authors
10:30	Some approaches for the field extrapolation in the context of inverse magnetisation problem	**Dmitry Ponomarev**
11:00	A tsunami inverse problem in the time domain: numerical analysis of an optimal control approach	Laurent Bourgeois, Philippe Moireau, **Raphaël Terrine**
11:30	Inversion-free binary segmentation for computed tomography and inverse scattering	**Jacob D. Rezac**, Zachary J. Grey, Newell H. Moser, Jason P. Killgore
12:00	Direct reconstruction for acoustic inverse Born scattering	**Lisa Schätzle**, Nuutti Hyvönen

T5 — Integral / DD / ABC Extras - Session Chair: Joseph Coyle

Time	Title	Authors
10:30	Quadrature for Space-Time Galerkin BEM of the Wave Equation	**Johannes Tausch**
11:00	Transcranial Ultrasound Simulations using Volume-Surface Integral Equations	**Elwin van 't Wout**, Alberto Almuna-Morales, Danilo Aballay, Ignacio Labarca-Figueroa, Pierre Gélat, Reza Haqshenas
11:30	Optimized Schwarz methods in time for discrete transport control	Duc Quang Bui, Bérangère Delourme, Laurence Halpern, **Felix Kwok**
12:00	A Multi-domain/Multi-method Preconditioner for the Electric Field Integral Equation	**Timothée Galtier**, David Levadoux

Lunch 12:30-14:00

Fri · Block B · 14:00 – 16:00

T1 — Domain Decomposition & Preconditioning - Session Chair: Lise-Marie Imbert-Gerard

Time	Title	Authors
14:90	Substructured Optimized Schwarz Domain Decomposition Solver on Multiple GPUs for Helmholtz Problems	**R. Greffe**, A. Chabib, A. Modave, C. Geuzaine
14:30	Domain Decomposition for Elastic Waves in Spatial Networks	**Joseph Holten**, Moritz Hauck
15:00	A Flux-Based Domain Decomposition Method for Time-Harmonic Wave Problems	**Nawfel Benatia**, Christophe Geuzaine
15:30	Accelerating WaveHoltz with Deflation	**Daniel Appelo**, William Henshaw

T2 — Full Waveform Inversion & Geophysical Applications - Session Chair: Allison Malcolm

Time	Title	Authors
14:00	Full Waveform Inversion Using Second Order Optimizers and Domain Decomposition with Spectral Coarse Space	**Boris Martin**, Christophe Geuzaine
14:30	An efficient numerical solver for non-periodic scattering problems in locally perturbed periodic structures	**Nasim Shafieeabyaneh**, Tilo Arens, Ruming Zhang
15:00	A Robust Model Order Reduction Method to Accelerate Full Waveform Inversion via Fréchet Derivatives	**Julien Besset**, Hélène Barucq, Victor Martins Gomes

T3 — Plasma & Mathematical Foundations - Session Chair: Daniel Appelo

Time	Title	Authors
14:00	Finite Element Formulation with Integral Dielectric Kernel for ICRH Plasma Heating on Non-Field-Aligned Meshes	**Jules Zaleski**, Bernard Reman, Philippe Lamalle, Christophe Geuzaine
14:30	Geometry-dependent well-posedness and finite elements for shear Alfvén waves	**M. Renoldner**, A. Buffa, T. Miehling, M. Picasso
15:00	A discontinuous Galerkin approach for the radiative transfer approximation of high-frequency vibrational energy flow on curved surfaces	**Robert J. Lockett**, David J. Chappell
15:30	Analysis and Simulation of 2D Maxwell's equations in Cold Plasma	**Manaswinee Bezbaruah1**, Patrick Ciarlet, Maryna Kachanovska

T4 — Time Integration & Nonlinear Waves - Session Chair: Tom Hagstrom

Time	Title	Authors
14:00	A Posteriori Estimates for Space Discretizations of the Wave Equation with Dynamic Meshes	Théophile Chaumont-Frelet, **Sumit Mahajan**, Martin Vohralík
14:30	Preserving multiple conserved quantities for the Korteweg–de Vries equation	Andy T.S. Wan, **Tareq Uz Zaman**
15:00	Nonlinear Acoustic Model Equations resulting from Non-Relativistic and Relativistic Lagrangian Pairing	**Markus Scholle**, Philip H. Gaskell, Meriem Al Ahmadi Hammou
15:30	A tsunami inverse problem in the time domain: numerical analysis of an optimal control approach	Laurent Bourgeois, **Philippe Moireau**, Raphaël Terrine